

RESULT NEC weights each error by a per-example cost $|\Delta|$ and reduces to error rate when costs are uniform. This reproduction confirms the paper's core finding on real public data: **NEC sits well below error rate across every obtainable modality**, and the paper's training-strategy negative result holds too — the entire reproduction ran on a single laptop for about \$0.

2.9×Jigsaw text
error / NEC**1.7×**Turkey image
error / NEC**1.4×**NHANES tabular
error / NEC**0.33** Δ -regression
MAE (log-odds)

1 The idea & the metric

Standard accuracy treats every mistake as equally bad, yet a **clear-cut** error (annotators agree 10–0) matters far more than an ambiguous **borderline** one. NEC weights each error by a per-example cost $|\Delta|$.

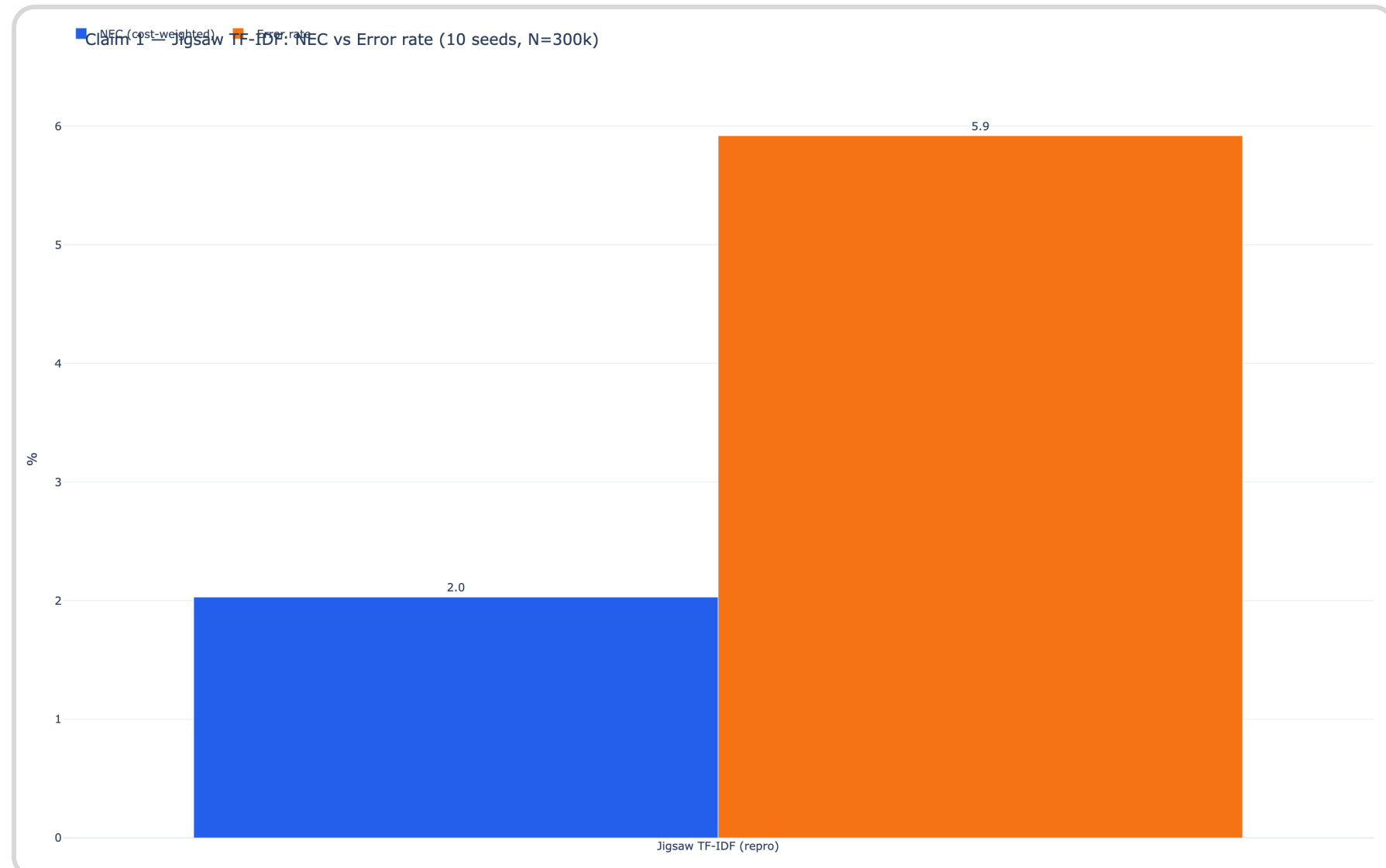
NORMALIZED EXCESS COST (EQ. 1)

$$NEC = \frac{\sum_i |\Delta_i| \mathbf{1}[\hat{y}_i \neq \text{sign}(\Delta_i)]}{\sum_i |\Delta_i|}$$

Uniform costs $\Rightarrow$ NEC = error rate. A high **error / NEC ratio** means mistakes fall on low-cost, ambiguous cases.

2 Claim 1 — Jigsaw: NEC $\approx$ 1/3 error

300k subsample of the real 1.8M comments; votes recovered (Eq. 3). TF-IDF + logistic regression, 10 seeds.

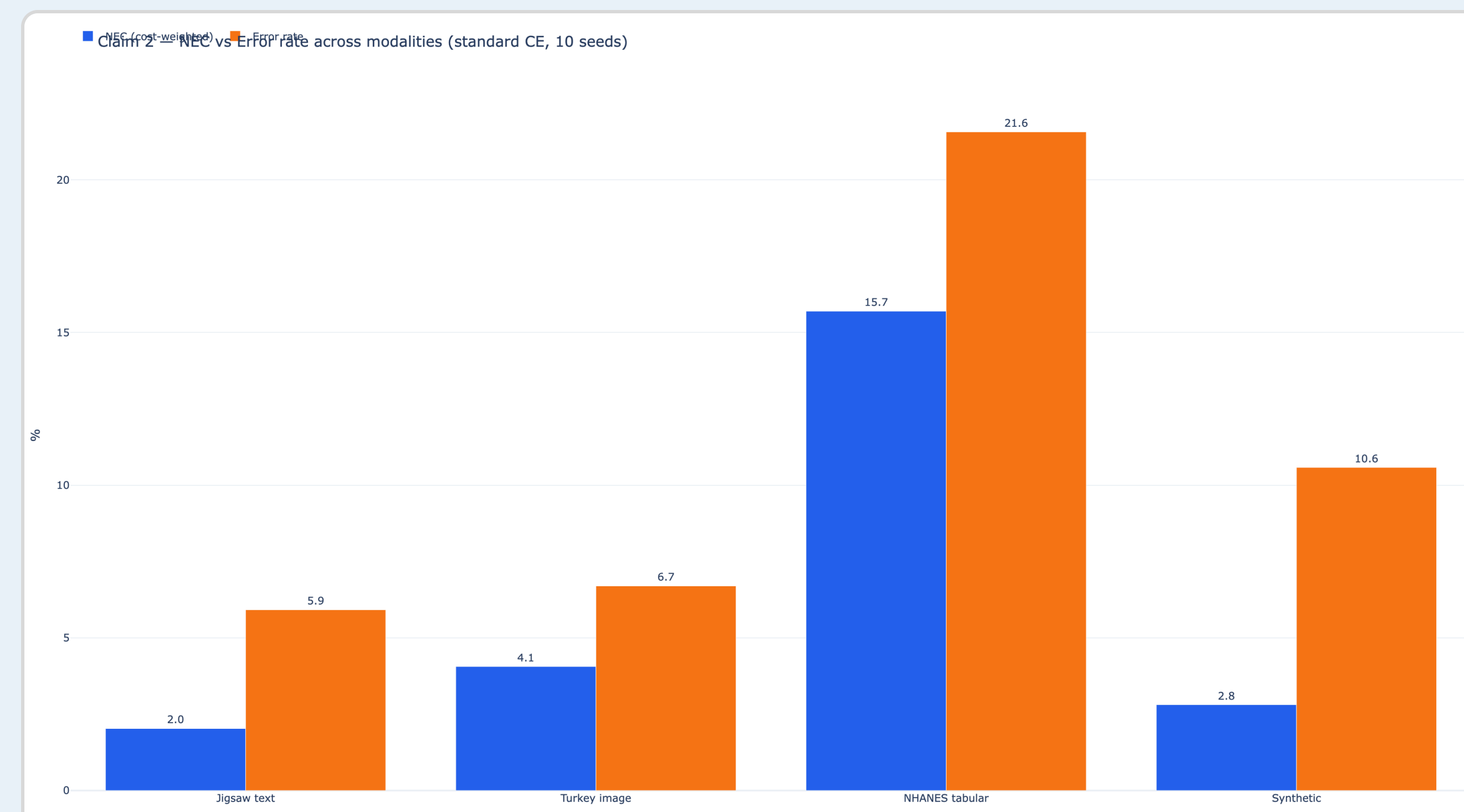


NEC 2.02% vs error 5.91% — error is **2.92×** higher (paper $\sim 3\times$).

2.02%NEC
(paper 1.76)**5.91%**error
(paper 5.34)**2.92×**ratio
(paper $\sim 3\times$)

3 Claim 2 — cross-modal divergence ★

Standard cross-entropy, 10 seeds, on the real public data. Every modality tested shows NEC materially below error rate, and the ordering of gap sizes matches the paper: text (many borderline cases) > image > tabular (few borderline cases).



Blue NEC, orange error rate. Gap largest on Jigsaw (many borderline comments), smallest on tabular.

Modality	NEC ↓	Error ↓	Ratio	Paper
Jigsaw text	2.02	5.91	2.92×	$\sim 3\times$
Turkey image	4.06	6.70	1.65×	1.6×
NHANES tab.	15.70	21.57	1.37×	1.5×
iNaturalist	—	—	blocked	1.1×

Turkey lands almost exactly on the paper (4.06 / 6.70 vs 4.16 / 6.68) under the any-injury binarization; NHANES reproduces from raw CDC blood-pressure files (N=7,457 vs 7,455). The gap ordering (text > image > tabular) tracks each dataset's share of borderline examples. **iNaturalist is data-blocked** — its Gemini annotations were never released.

4 Claim 3 — three cost sources

All three derivations implemented and verified against the equations; the Jigsaw Δ recomputed from stored votes matches the recovered values exactly.

EQ. 3 — MULTI-ANNOTATOR VOTES

$$\Delta_i = \log \frac{n_{\text{yes}} + 1}{n_{\text{no}} + 1}$$

EQ. 4 — THRESHOLD DISTANCE

$$\Delta_i = z_i - \tau$$

Third source: direct ratings (Δ = score – midpoint) — synthetic demo; iNat ratings unreleased. Real Fig. 2 panels reproduce.



Red line marks the $\Delta=0$ decision boundary; the shape of each cost distribution drives the size of its error/NEC gap.

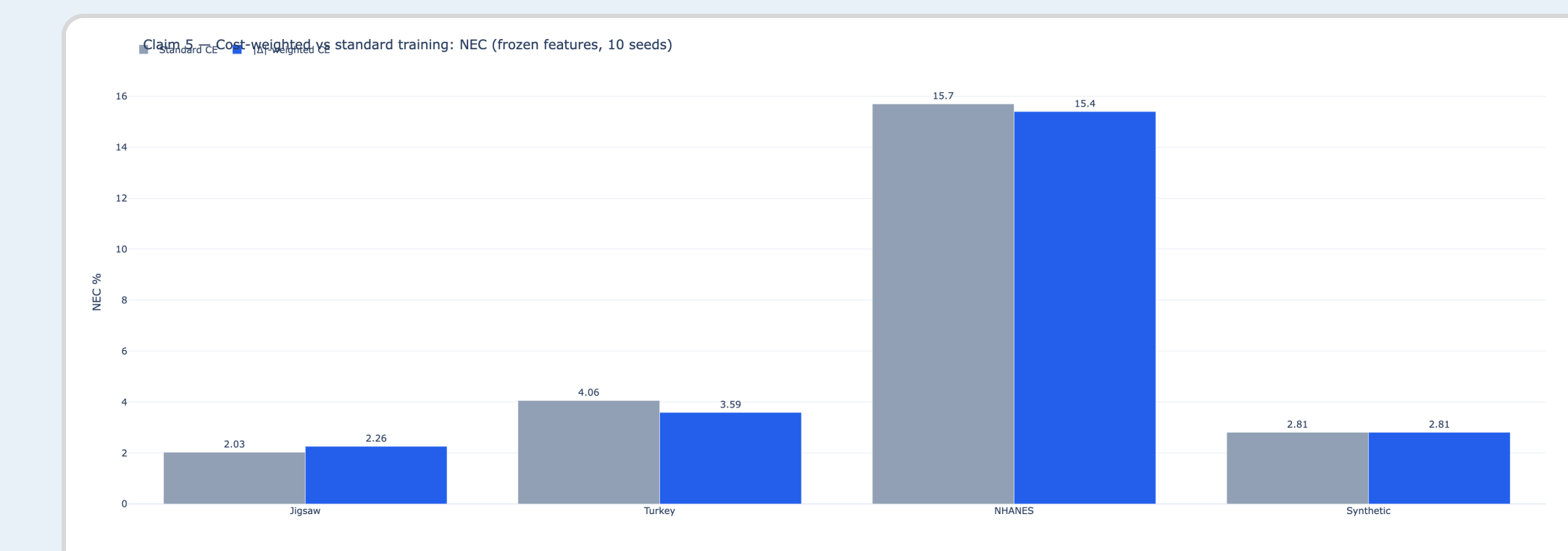
5 Claim 6 — Δ -regression = calibration

Δ -regression slightly improves Jigsaw NEC (1.96% vs 2.02%; paired Δ -0.06 pts, significant) and adds a continuous margin diagnostic.

0.325Jigsaw Δ -MAE
(paper 0.30)**1.96**reg NEC
(class 2.02)**-0.06**paired Δ NEC pts
(Jigsaw, sig.)

6 Claim 5 — cost-weighting is inconsistent

Relative NEC change, $|\Delta|$ -weighted vs standard CE — inconsistency holds; Jigsaw diverges (harms).



Blue = $|\Delta|$ -weighted, grey = standard CE. Cost-weighting helps on Turkey but is neutral or harmful elsewhere.

Dataset	Δ NEC (repro)	Paper
Turkey	-11.6%	-14.5%
NHANES	-1.9%	-2.5%
Jigsaw	+11.4%	~ 0

The \$5.3 **feature-predictability** mechanism fits the real-data ranking, but the synthetic test built to isolate it stayed neutral, so the mechanism remains unconfirmed here.

7 Scope, blockers & cost

- 3 of 4 datasets public (Jigsaw, Turkey, NHANES), all verified by download.
- iNaturalist: data-blocked — the Gemini 7-point ratings were never released.
- Fine-tuned Table 2 (Claim 4): not reproduced — no GPU credits (HF Jobs 402); the MPS toy is exploratory only.

1 × M4 · ~ 1.2 h · \$0 cloud. Claims 1–2 full; 3/5/6 partial.

8 Takeaways

IDEA. Weight errors by disagreement-derived cost $|\Delta|$; NEC exposes model quality error rate hides.

RESULT. NEC $\ll$ error rate reproduces on text, image & tabular (2.9×, 1.7×, 1.4×).

TRAINING. Cost-weighting is inconsistent as claimed — helps Turkey, harms Jigsaw; mechanism unconfirmed.

HONESTY. iNaturalist data-blocked; fine-tuning blocked, no GPU credits (402) — reported, not proxied.